

Practical Week – 2

Task : 1

```
#include <stdio.h>

#include <stdlib.h>

#include <unistd.h>

#include <sys/wait.h>

#include <string.h>


#define MAX_INPUT 100


int main()
{
    char input[MAX_INPUT];
    char *args[MAX_INPUT];
    int i = 0;
    pid_t pid;
    int status;


    printf("Enter a Linux command: ");
    fgets(input, sizeof(input), stdin);


    // Remove newline
    input[strcspn(input, "\n")] = '\0';


    // Convert input into arguments
    char *token = strtok(input, " ");


    while (token != NULL)
    {
```

```

        args[i++] = token;
token = strtok(NULL, " ");
    }

    args[i] = NULL;

    // Create child process
pid = fork();

    if (pid < 0)
    {
        perror("fork failed");
return 1;
    }

    if (pid == 0)
    {
        // Child process        printf("\nChild
Process\n");    printf("Parent PID : %d\n",
getppid());    printf("Child PID : %d\n",
getpid());

        printf("\nExecuting command: %s\n", args[0]);

        // Execute command
execvp(args[0], args);

        // Only reached if execvp fails
perror("execvp        failed");
exit(1);

```

```
    }  
else  
{  
    // Parent process  
    printf("\nParent Process\n");  
    printf("Parent PID : %d\n", getpid());  
    printf("Child PID : %d\n", pid);  
  
    // Wait for child  
    wait(&status);  
  
    printf("\nChild process completed.\n");  
}  
  
return 0;  
}
```

Output :

```
anuan@anushka:~$ nano command_executor.c
anuan@anushka:~$ gcc command_executor.c -o command_executor
anuan@anushka:~$ ./command_executor
Enter a Linux command: ls

Parent Process
Parent PID : 492
Child PID  : 493

Child Process
Parent PID : 492
Child PID  : 493

Executing command: ls
File1.txt      command_executor  keyboard_input      parser_quotes.c
File2.txt      command_executor.c keyboard_input.c     process
LICENSE        docs              main_loop           process.c
Makefile       file_copy         main_loop.c         process.c.save
OSSP           file_copy.c       obj                 scripts
README.md      fork_exec         parser_double_quotes src
assets         fork_exec.c       parser_double_quotes.c tests
bin            include           parser_quotes
```

Child process completed.

```
anuan@anushka:~$
```

Task : 2

```

anuan@anushka:~$ nano command_executor.c
anuan@anushka:~$ gcc command_executor.c -o command_executor
anuan@anushka:~$ ./command_executor
Enter a Linux command: ls

Parent Process
Parent PID : 492
Child PID  : 493

Child Process
Parent PID : 492
Child PID  : 493

Executing command: ls
File1.txt  command_executor  keyboard_input  parser_quotes.c
File2.txt  command_executor.c keyboard_input.c process
LICENSE    docs                main_loop      process.c
Makefile   file_copy           main_loop.c    process.c.save
OSDP       file_copy.c         obj             scripts
README.md  fork_exec           parser_double_quotes src
assets     fork_exec.c         parser_double_quotes.c tests
bin        include             parser_quotes

Child process completed.
anuan@anushka:~$
anuan@anushka:~$ uname -a
Linux Anushka 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 19 07:09:16 UTC 2026 x86_64 GNU/Linux
anuan@anushka:~$ lscpu | head -20
Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:                39 bits physical, 48 bits virtual
Byte Order:                  Little Endian
CPU(s):                       16
On-line CPU(s) list:         0-15
Vendor ID:                    GenuineIntel
Model name:                   13th Gen Intel(R) Core(TM) i7-13620H
CPU family:                   6
Model:                        186
Thread(s) per core:           2
Core(s) per socket:           8
Socket(s):                     1
Stepping:                     2
BogoMIPS:                     5836.80
Flags:                         fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush
                                mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtscp lm constant_tsc rep_good nopl
                                VT-x
Virtualization:               VT-x
Hypervisor vendor:            Microsoft
anuan@anushka:~$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root         1      0  0 07:00 ?        00:00:00 /sbin/init
root         2      0  0 07:00 ?        00:00:00 /init
root         6      2  0 07:00 ?        00:00:00 /usr/lib/systemd/systemd --switched-root --system --deserialize 22
root        46      1  0 07:00 ?        00:00:00 /usr/lib/systemd/systemd-journald
root        88      1  0 07:00 ?        00:00:00 /usr/lib/systemd/systemd-udev
message+   109      1  0 07:00 ?        00:00:00 /usr/bin/dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation
root       319    316  0 07:01 pts/0    00:00:00 -bash
root       570      2  0 07:01 ?        00:00:00 /init
root       625    573  0 07:02 pts/2    00:00:00 ps -ef
root       626    319  0 07:02 pts/0    00:00:00 ps -ef
anuan@anushka:~$ top -b -n 1 | head -20
top - 07:02:44 up 2 min, 1 user, load average: 0.00, 0.01, 0.00
Tasks: 29 total, 1 running, 28 sleeping, 0 stopped, 0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni, 99.8 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 7892.0 total, 6320.8 free,  601.5 used,  969.7 buff/cache
MiB Swap: 2048.0 total, 2048.0 free,  0.0 used, 7290.5 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
 318 root        20   0   24160   1220   11360 S   0.0   0.2   0:00.11 /init
   2 root        20   0   24160   1220   11360 S   0.0   0.2   0:00.02 /sbin/init
   6 root        20  -10  23552    624    4224 S   0.0   0.1   0:00.00 /init
 219 root        20   0   18212    320    2320 S   0.0   0.0   0:00.00 -bash
 625 root        20   0    9100    380    2884 R   0.0   0.0   0:00.00 ps

```